

**COMPREHENDING LEVELS OF EXPLAINABILITY AND AI
RECOMMENDATIONS (CLEAR) STUDY SERIES**

Virginia Sullivan

A DISSERTATION

**Submitted in partial fulfillment of the requirements
for the degree of Doctor of Philosophy
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Abstract

COMPREHENDING LEVELS OF EXPLAINABILITY AND AI RECOMMENDATIONS (CLEAR) STUDY SERIES

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**A dissertation submitted in partial fulfillment of the requirements
for the degree of Doctor of Philosophy**

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Artificial Intelligence (AI) systems can offer a great deal to improve decision-making through efficiency and predictive accuracy due to their unique ability to rapidly analyze large amounts of data relevant to the decision. Yet many AI systems remain opaque, limiting users' ability to evaluate outputs and leading to issues of mistrust or over-reliance. This dissertation investigates how varying levels of AI explainability affect trust, perceived reliability, understanding, and confidence, and identifies perceptual and behavioral indicators of over-reliance behavior in AI-assisted decision-making scenarios. This dissertation is organized as a three-study series. The CLEAR survey study uses a mixed design to examine the impact of explanation depth on trust-related outcomes. Participants engage with AI-assisted decision scenarios and respond to measures of trust, perceived reliability, confidence in the accuracy of the AI, and understanding of the explanation. Scenario context is also evaluated as a potential moderator. Study 2 builds on this by constructing an Over-Reliance Index (ORI), a combined measure of potential over-reliance built on user perceptions that are related to reliance behavior

in previous literature. Using data from the CLEAR survey, this study identifies dispositional and contextual predictors of over-reliance and employs regression analysis to examine differences in reliance behaviors by explanation complexity level and cluster analysis to reveal user profiles. The CLEAR-Engage study shifts from self-report to decision behavior by examining responses to AI recommendations within a simulated, screenshot-based decision-making task. Participants completed a hostage-rescue task in which they were presented with varying levels of AI support. Behavioral measures (*e.g.*, agreement with AI recommendations and detection of AI errors), along with self-report measures of trust and understanding, were used to validate the ORI and identify behavioral markers of over-reliance. Together, these studies aim to advance understanding of explainability's role in trust calibration, establish behavioral measures of over-reliance, and inform the design of user-centered AI systems that support effective human-AI teaming in a variety of contexts.

I have followed the approved guidelines of my graduate program and the UAH Graduate School for the use of generative Artificial Intelligence (AI) in the writing of this dissertation. The following generative AI tools were used for the associated purpose(s) listed after each tool:

ChatGPT: Scenario generation in the CLEAR survey study. Revising, editing, and proofreading text for clarity and coherence.

PREVIEW

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Completing this dissertation required more resilience and growth than I anticipated, and I am proud of the perseverance it represents. Lastly, I would like to thank my dogs, Eevee and Jack, for the comfort, joy, and anxiety-relieving cuddles that carried me through this process.

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PREVIEW

*It is hoped that by trying to imitate the behavior of a human brain (or that of some other animal) by means of an electronic device—or by failing to do so—one may learn something of importance concerning the brain's workings. Finally, there is the optimistic hope that for similar reasons AI might have something to say about deep questions of philosophy, by providing insights into the meaning of the concept of **mind**.*

– Roger Penrose, *The Emperor's New Mind* (1989)

PREVIEW

Chapter 1. Introduction

The purpose of this dissertation is to investigate the relationship between Explainable Artificial Intelligence (XAI) complexity and user perceptions (trust, understanding, perceived reliability, and accuracy confidence), as well as factors that lead to over-reliance and over-trust in artificial intelligence (AI), despite explanation complexity. More specifically, this research seeks to identify the conditions under which users accurately rely on recommendations from AI versus when they over-rely (accepting AI output without sufficient understanding or engagement). Across three interrelated studies, this project aims to determine the optimal level of explainability for calibrating trust and reliability, develop a composite over-reliance index (ORI) based on perceptual and behavioral indicators, and validate the ORI through tasks designed to mimic real-world decision-making contexts. The findings will contribute to a more nuanced understanding of XAI in human-AI interaction and inform XAI design that supports effective and informed decision-making. Overall, this research aims to examine how different AI explanations influence user perceptions and reliance behaviors across a variety of decision-making contexts.

1.1 Artificial Intelligence

This dissertation closely examines user behaviors during their interactions with AI systems. The idea of AI computational systems was conceived in the late 1940s and early 1950s, and the idea was first fully conceptualized by Alan Turing (1950) when he asked the question, “Can machines think?” in his landmark article, “Computing Machinery and Intelligence.” In

1956, scientists at Dartmouth University were inspired by Turing and the possibilities of artificial intelligence and began a 6-week summer workshop aimed at creating machines capable of outsmarting humans (Bostrom, 2014). Over the next six decades, progress in the field of artificial intelligence ebbed and flowed and coincidingly investor interest and disinterest. A major change in the field occurred in 2012 when Krizhevsky *et al.* (2012) published results of a deep convolutional neural network (CNN) that vastly outperformed any prior models at visual recognition. The findings of Krizhevsky *et al.* showed the power that deep learning could have to transform AI and inspired applications of deep learning techniques across various other fields.

A second landmark finding came in 2014 when Goodfellow *et al.* introduced generative adversarial nets (GANs), which use two neural networks to generate realistic text and images. These seminal findings led to a boom in AI research and advancements in a wide range of fields. In healthcare, a CNN was trained to identify skin cancer at a level comparable to dermatologists (Esteva *et al.*, 2017). Large language models (LLMs) were developed based on findings by Vaswani *et al.* (2017) that improved performance and reduced training times. The AI systems that resulted from these findings quickly became mainstream and have already had a profound impact on the lives of many individuals and organizations. For example, a study on automatization of jobs examined 702 occupations and found that around 47% of jobs in the US are at a high risk of being automated within the next decade or two (Frey & Osborne, 2017). Other researchers have examined the likely effects of AI on security, transportation, healthcare, entertainment, and multiple other areas of life (Brundage *et al.*, 2018; Dul, 2022; Mollick, 2024; Stone *et al.*, 2022).

These advances in AI technology have made it clear that AI systems have a broad range of uses and applications. One particular advantage of AI that increases its usefulness in many of

these applications is its ability to quickly and effectively make decisions and provide information to aid in the decision-making process. AI can offer a great deal to improved decision-making through efficiency and predictive accuracy, due to its unique ability to rapidly analyze large amounts of data relevant to a decision. The human brain also has the ability to process and analyze data relevant to a decision, but there are limitations to the amount of data the brain is able to process, especially in time critical situations (C. D. Wickens *et al.*, 2015). The brain often fills in gaps in knowledge and processing ability by using mental shortcuts (*e.g.*, gestalt organizational principles and heuristics) and will often attend to only salient, expected, or high value information (Kahneman, 2011; Proctor & Van Zandt, 2018; J. R. Wickens *et al.*, 2003). AI excels where the human brain falls short due to its ability to rapidly process vast amounts of information relevant to a decision.

However, there are challenges associated with the use of AI in decision-making. Informed decision making requires consideration of shared understanding, user cognitive load, automation bias, user trust, perceived reliability of the AI, and user confidence in the decision, among other cognitive and computing considerations. It can be difficult for the user to feel their decision is informed and well-made if the AI system is unable to provide information and explanations that allow for shared understanding of context, consider cognitive load, and give the information necessary for trust calibration and perceived reliability. Many AI systems operate as so-called “black boxes,” where the logic behind decisions remains opaque even to developers, let alone end-users. In this case an explanation describing how the system arrived at a decision can provide needed context to the user.

1.2 Explainable Artificial Intelligence

One proposed solution to the challenges associated with human-AI decision making and the need for explanation is the use of XAI. XAI refers to a broad class of approaches designed to make the decision-making process of AI systems more transparent, interpretable, and understandable to human users (Ghela *et al.*, 2024). The field of XAI has received a great deal of attention in recent years but remains in a state of flux. There is disagreement among computer scientists, human-computer interaction (HCI) researchers, and psychologists about how to best implement and visualize explanations to the user. The type and delivery method of explanations should be context and function dependent due to the wide range of AI applications, but it must also be trustworthy, understandable, and useful to the user, regardless of context or function. Therefore, research must be done to measure user perceptions of AI that will aid in the design of effective XAI systems.

Initially, XAI implementation largely relied on the assumption that increasing the amount of explanations leads to enhanced user trust and understanding, as providing detailed explanations can help users evaluate AI recommendations (Druce *et al.*, 2021; Othman, 2025; Walzl & Vogl, 2018). However, other research suggests that user perceptions have an inverted-U relationship with explanation complexity (Abdul *et al.*, 2020; Ghai *et al.*, 2021; Walmsley, 2021). At high levels of complexity, some studies have shown that explanations begin to have a negative impact on user perceptions (Abdul *et al.*, 2020; Ghai *et al.*, 2021; Walmsley, 2021). Explanations that are too robust or overly complex can increase cognitive load, overwhelm users, or paradoxically increase over-reliance on the AI system by creating a false sense of understanding (Abdul *et al.*, 2020; Ghai *et al.*, 2021; Ngo, 2025; Walmsley, 2021).

Alternatively, the absence of explanations or explanations that are too simple may fail to convey essential decision-making information, leaving users unable to judge the system's reliability (Choung *et al.*, 2023; Parasuraman & Riley, 1997a). Ideally, users are able to obtain the exact amount of information they need to reach a decision that is based on accurately calibrated trust and perceived reliability of the system.

This dissertation addresses the critical need for user sensitive XAI frameworks that promote calibrated trust and reliability based on appropriate user understanding and confidence to inform user decision-making. Building on existing research in human factors and cognitive psychology, this study series aims to examine how different levels of explanation complexity affect user perceptions of trust, confidence, understanding, and perceived reliability, while also identifying when those effects contribute to over-reliance behavior in human-AI decision making interactions.

1.3 Organization of the Dissertation

This dissertation is organized into eight chapters that together build the Comprehending Levels of Explainability and AI Recommendations (CLEAR) study series. The series integrates three primary research objectives that examine how varying levels of explainability influence user trust and reliance on AI systems.

Chapter 1 introduces the overarching topic, establishing the context of AI, XAI, and the psychological foundations that motivate this research. The chapter concludes by outlining the structure and objectives of the clear study series.

Chapter 2 presents a comprehensive literature review that bridges the computer science and psychological perspectives of XAI. The chapter reviews relevant terminology and theoretical

frameworks that inform the design of the CLEAR study series and identifies critical gaps this research aims to address.

Chapter 3 covers research objective 1, the CLEAR survey. The chapter begins by defining the specific research questions and hypotheses associated with the CLEAR survey. The overall objective of this study is to: determine the optimal amount of explanation complexity (low, medium, high) that leads to accurately calibrated trust, reliance, understanding, and confidence across a variety of decision-making scenarios, while examining variations in these effects across contexts (scenarios). Methodology, analysis approach, results, and discussion for the CLEAR survey are then outlined.

Chapter 4 covers research objective 2, the ORI and identifying predictors of over-reliance behavior. The chapter begins by defining the specific research questions and hypotheses for the ORI study. The overall objective of this study is to: identify when and why over-reliance on AI occurs by developing a novel index (ORI) and to determine the impact of XAI on over-reliance. Methodology, analysis approach, results, and discussion for the ORI study are then outlined.

Chapter 5 covers research objective 3, the CLEAR-Engage study. The chapter begins by defining the specific research questions and hypotheses for the CLEAR-Engage study. The overall objective of this study is to: validate the ORI using behavioral measures collected during a simulated AI-assisted decision-making task, identify behavioral markers of over-reliance, and examine how varying levels of explanation complexity influence reliance on AI recommendations. Methodology, analysis approach, results, and discussion for the CLEAR-Engage study are then outlined.

Chapter 6 covers the overall discussion for the implications of the CLEAR study series as a whole and integrates findings across the three studies to discuss theoretical, methodological,

and practical implications for XAI design, trust calibration, and human-AI interaction.

Limitations are also addressed. Chapter 7 outlines the intellectual merit and broader impacts of the research, situating its contributions within both theoretical and applied domains of Human-AI Interaction and human factors psychology.

Chapter 8 concludes the dissertation by synthesizing contributions across the three studies and articulating the cumulative impact of the CLEAR study series.

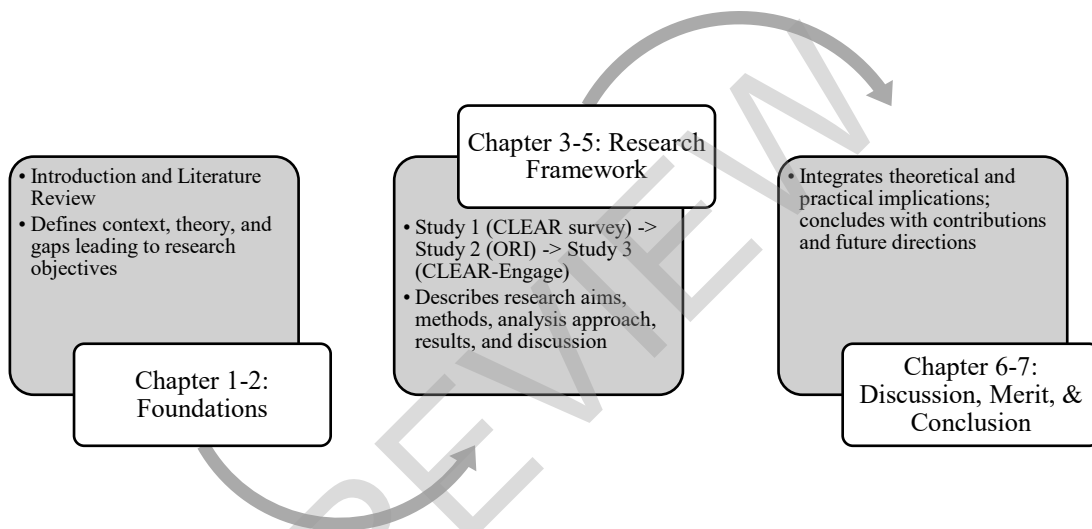


Figure 1.1 Organization of the Dissertation and Study Flow.

Chapter 2. Literature Review

This review examines the current state of XAI from a human-centered perspective, with a focus on how explanations are perceived, processed, and used in decision-making contexts. Drawing on research from psychology, human factors, and human-AI interaction, the review synthesizes findings related to cognitive load theory (CLT), trust in technology, automation bias, and AI reliance behaviors, and identifies gaps in our understanding of how explanation design influences user trust calibration and reliance behavior.

2.1 Definitions and Terminology

First, a preliminary overview of terms related to XAI research is provided. Inherently, AI systems are typically “black box” systems, meaning that the inner workings of the system are completely opaque. Ideally, XAI allows the inner workings of the system to become more transparent to the user, sometimes referred to as “white box” AI. The term black box is not unique to AI and has been used widely to describe what happens in between the input and output of a system when the internal structure is unknown. In behavioral psychology, the mind can be thought of as a black box because what is happening between the stimulus input and behavioral output/reaction cannot be directly observed (Skinner, 1987). Neural network AI systems are modeled on the brain and diffuse stored information in several complex networks that are incredibly difficult to decipher, which makes their inner workings a black box, much like the human mind (Castelvecchi, 2016). Neural network algorithms are exceptionally complex black